

$$f(x, y) = x \sin y + y \cos x$$

$$\frac{\partial f(x, y)}{\partial x} = \sin y + y(-\sin x)$$

$$= \sin y - y \sin x$$

$$\frac{\partial f}{\partial y} = x \cos y + \cos x$$

Construct C.G.

$$f(x, y) = t_1 + t_2$$

$$t_1 = x * t_3$$

$$t_3 = \sin(y)$$

$$t_2 = y * t_4$$

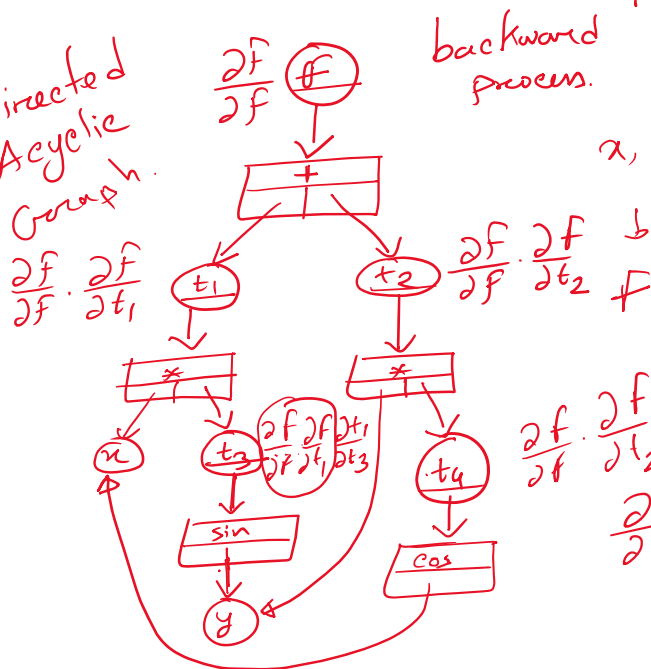
$$t_4 = \cos(x)$$

operand
variables
 t, f, x, y

operators
 $+, \sin, \cos, *$

Autodiff

Directed
Acyclic
Graph



backward
process.

$$\frac{\partial}{\partial x}, \frac{\partial}{\partial y}$$

x, y

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial t_1} \frac{\partial t_1}{\partial t_3} \frac{\partial t_3}{\partial y} + \frac{\partial f}{\partial t_2} \frac{\partial t_2}{\partial y}$$

$$= 1 \cdot 1 \cdot x \cos y + 1 \cdot 1 \cos x$$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial t_1} \frac{\partial t_1}{\partial x} + \frac{\partial f}{\partial t_2} \frac{\partial t_2}{\partial x}$$

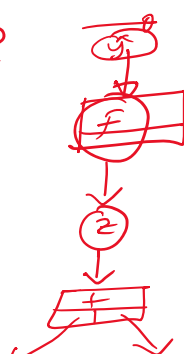
$$= 1 \cdot 1 \cdot \sin y + 1 \cdot 1 y(-\sin x)$$

$$= \sin y - y \sin x$$

$\underline{x}, \underline{w}, \underline{b}$

$$z = wx + b$$

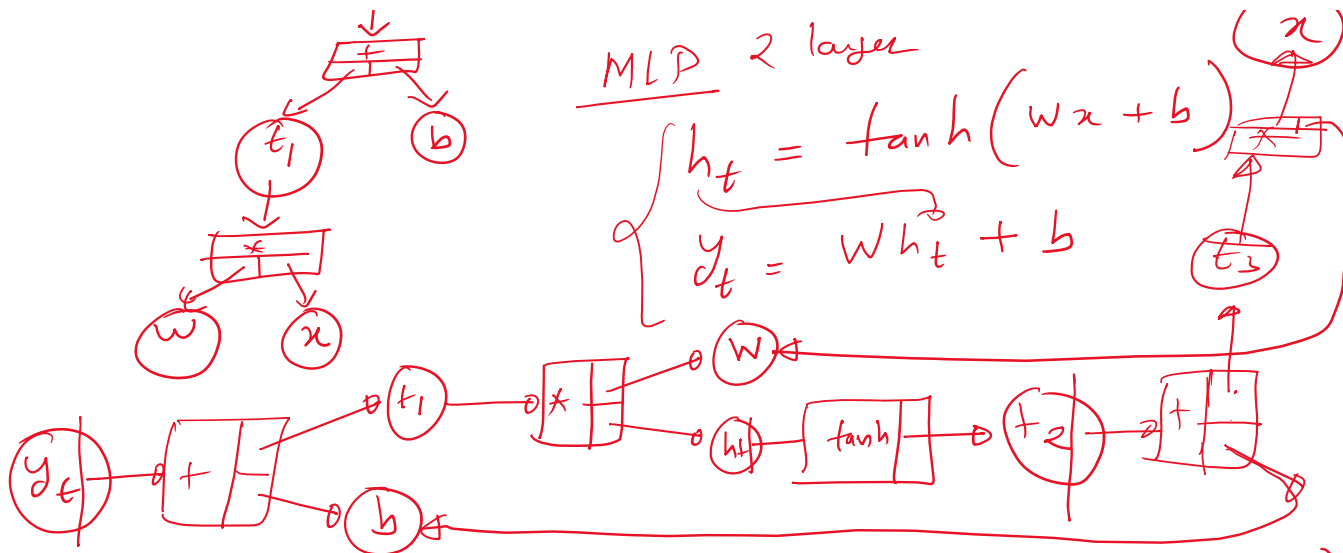
$$\hat{y} = f(z)$$



$$\frac{\partial}{\partial w} \frac{\partial}{\partial x} \frac{\partial}{\partial b}$$

MLP 2 layer





$$f(x, y, z) = \frac{xz}{z} + \sin(yx) + \cos(x^2 - z^2)$$

$$\frac{\partial f}{\partial x} = 1 + y \cos(yx) - 2xz \sin(x^2 - z^2)$$

$$\frac{\partial f}{\partial z}$$

$$t_1 = x * z$$

$$t_2 = y * x$$

$$t_3 = x * x$$

$$t_4 = z * z$$

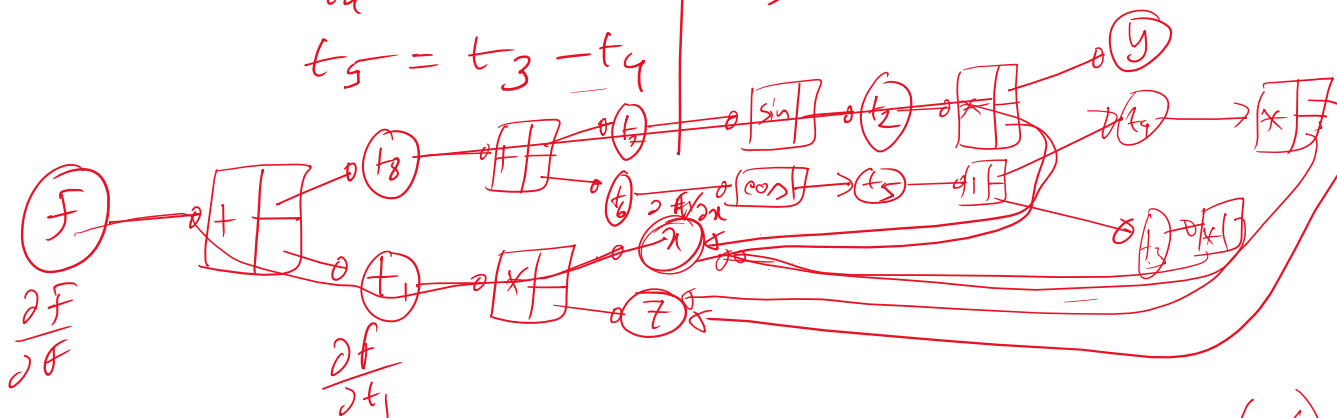
$$t_5 = t_3 - t_4$$

$$t_6 = \cos(t_5)$$

$$t_7 = \sin(t_2)$$

$$t_8 = t_7 + t_6$$

$$t_9 = t_1 + t_8$$



$$z = x * x$$

$$y = x * x$$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial t_1} \frac{\partial t_1}{\partial x} + \frac{\partial f}{\partial t_8} \frac{\partial t_8}{\partial t_6} \frac{\partial t_6}{\partial t_5} \frac{\partial t_5}{\partial t_3} \frac{\partial t_3}{\partial x}$$

$$= 1 \cdot 1 \cdot z + 1 \cdot 1 \cdot 1 \cdot (-\sin(x^2 - z^2)) \cdot 1 \cdot (2x)$$

$$= z - \sin(x^2 - z^2) + \dots$$